

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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Flow

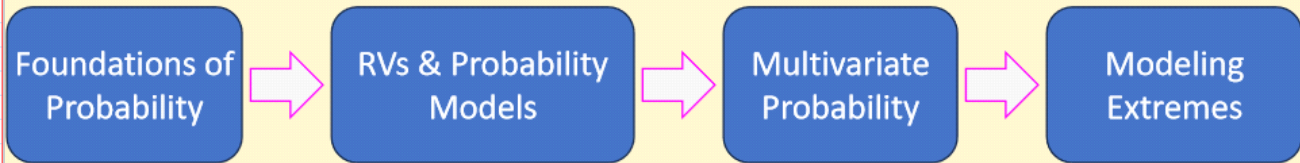
- Announcement
- Recap
- Module-2 starts

Announcement

- Lectures
 - No class on next Tuesday (29th Sep)
 - Class tomorrow (Friday 25th Sep) at 10 am
- Check GitHub regularly

Course Flow

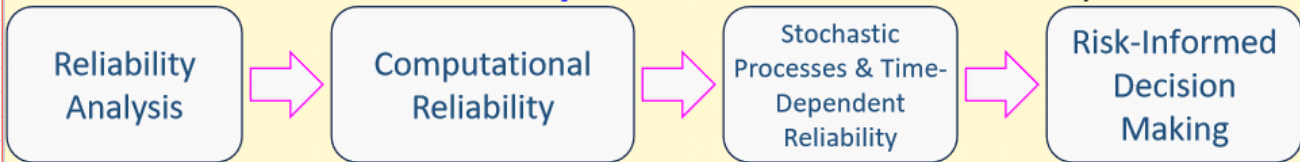
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data



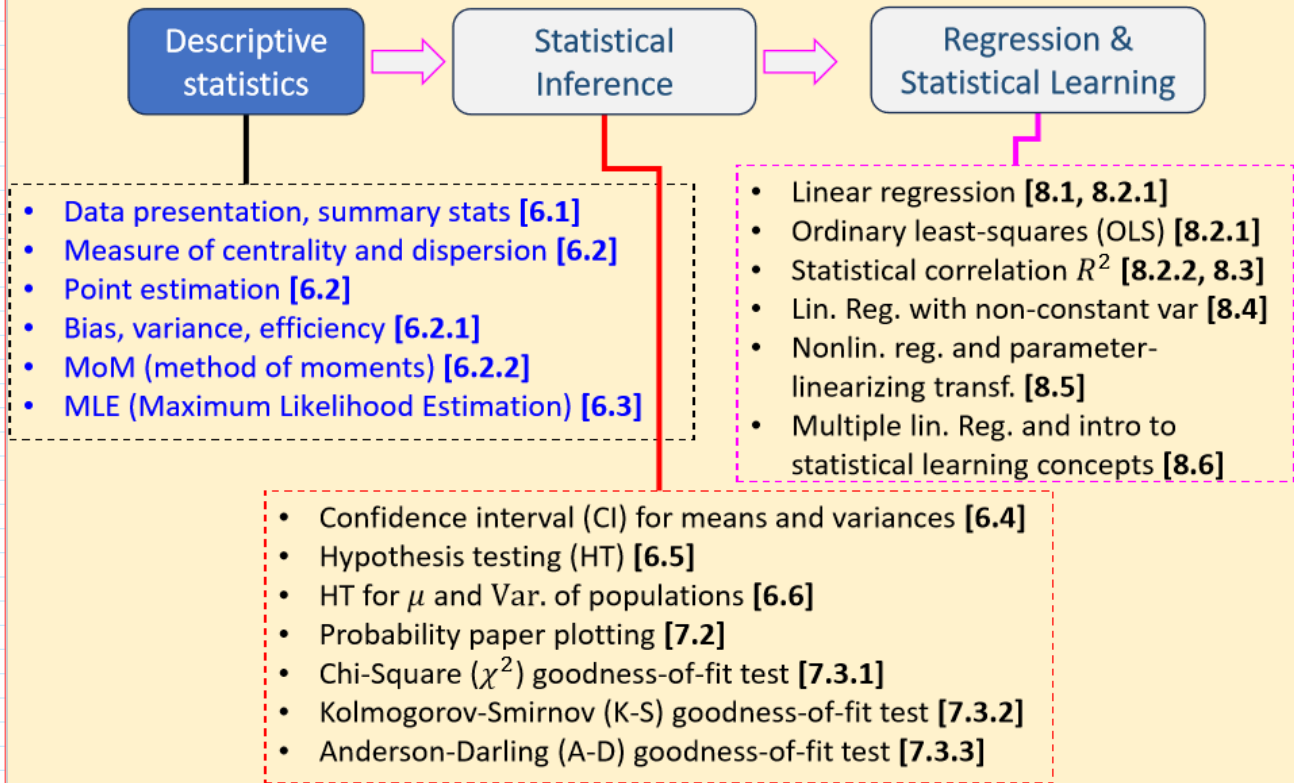
Module 3: **Reliability** – Decision under Uncertainty



L26
Sep 24, 2026

Module II: Chapter 1 (M2C1)

Module 2: **Statistics** – Learning from Data



M2C1

Point Estimation Framework: $\hat{\theta}$ to characterize unknown population parameters from finite empirical observations.

Properties of Estimators: unbiasedness, efficiency, consistency, and variance bounds to evaluate optimal estimator trade-offs.

Method of Moments (MoM): parameter estimates by equating theoretical population moments to corresponding sample moments.

Maximum Likelihood Estimation (MLE): construct likelihood and log-likelihood functions to estimate parameter values that maximize the probability of observed sample data.

Sampling Distributions & Standard Errors: Quantify the variability of point statistics.

Motivation

How to extract single best-estimate model parameters from noisy, finite field measurements?

Bias and precision

N,T: How close the dart throws are to the bullseye?

• Accuracy/bias:

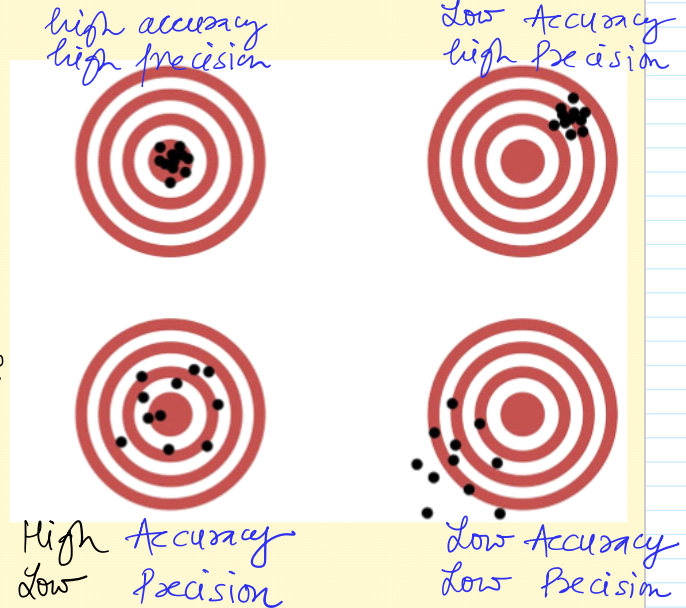
→ closeness to center, on an average

→ Neutrality/Unbiasedness

• Precision:

How close are the dart throws to each other?

→ Consistency/ Repeatability



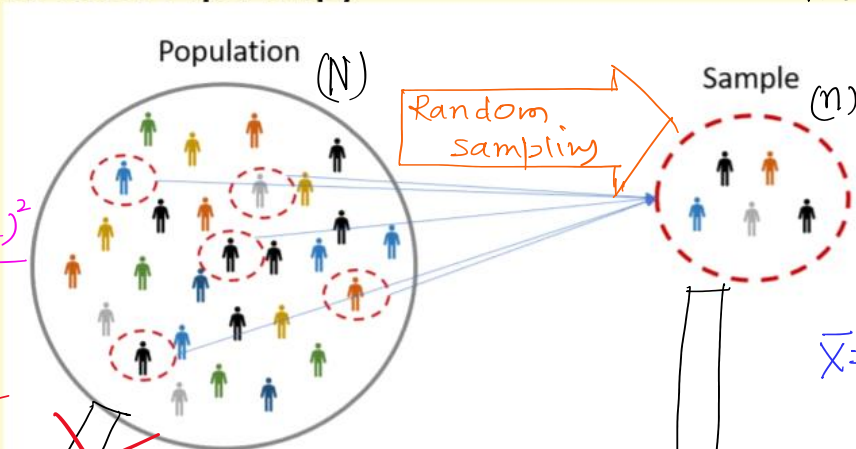
8

Random Sampling

e.g., Election polls, Mechanical Properties

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$



It cannot be observed directly!

POPULATION STATISTIC

INFERENCE

SAMPLE STATISTIC

$$\bar{X} = \hat{\mu} = \frac{\sum_{i=1}^n x_i}{n}$$

$$se = \hat{\sigma}^2 = \frac{\sum_{i=1}^n (x_i - \hat{\mu})^2}{n-1}$$

Parameter estimation: method of moments

Given: 1) Data & Probability Model, to estimate Parameters of the

- Obtain the moments (e.g., mean, variance) from two sources:

- Observations:

$$\hat{\mu} = E[X] = \left\{ \begin{array}{l} \text{mean} \\ \text{stdev} \end{array} \right\} \text{ of observed data}$$

$$\hat{\sigma}^2 = E[(X - \mu)^2] =$$

- Parameters:

eqn. for $\left\{ \begin{array}{l} \text{mean} \\ \text{stdev} \end{array} \right\}$ of a PDF

For point-estimate,

- Equate them to solve for the parameters

Distributions and parameters

Dist.	PDF	CDF	Mean & Variance
Normal/ Gaussian	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$	$F(x) = \Phi(x)$ Use z-table	$E[X] = \mu$ $\text{Var}[X] = \sigma^2$
Uniform	$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$	$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{otherwise} \end{cases}$	$E[X] = \frac{1}{2}(a+b)$ $\text{Var}[X] = \frac{1}{12}(b-a)^2$
Exponential	$f(x) = \lambda \exp(-\lambda x)$	$F(x) = 1 - \exp(-\lambda x)$	$E[X] = \frac{1}{\lambda}$ $\text{Var}[X] = \frac{1}{\lambda^2}$
Gumbel	$f(x) = \frac{1}{\beta} \exp[-z + \exp(-z)],$ where $z = \frac{x-\alpha}{\beta}$	$F(x) = \exp[-\exp(-z)]$	$E[X] = \alpha + \beta\gamma$ $\text{Var}[X] = \frac{\pi^2}{6}\beta^2$ $\gamma \approx 0.577$
Lognormal	$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2\right]$	$F(x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$ Use z-table	$E[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right)$ $\text{Var}[X] = [\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$

Example

- Assume that Earthquakes in Rome expressed as MCS index follow Gumbel type-I distribution. Earthquake intensity and number of earthquakes between 1000 CE – 1980 CE are:

MCS intensity, x_i	2	3	4	5	6	7	$\sum x = \sum x_i f_i$	$\sum x^2 = \sum x_i^2 f_i$	$\sum f_i$
Number, f_i	113	132	56	22	4	2	994	3328	329

Find the parameters of the distribution.

- Observations:

$$E[X] = \mu_X = 994/329 \approx 3.02$$

$$Var[X] = E[X^2] - (E[X])^2 \approx 0.99$$

- Parameters for Gumbel type-I:

$$E[\cdot] = \alpha + \beta\gamma$$

$$Var[\cdot] = \frac{\pi^2}{6}\beta^2$$

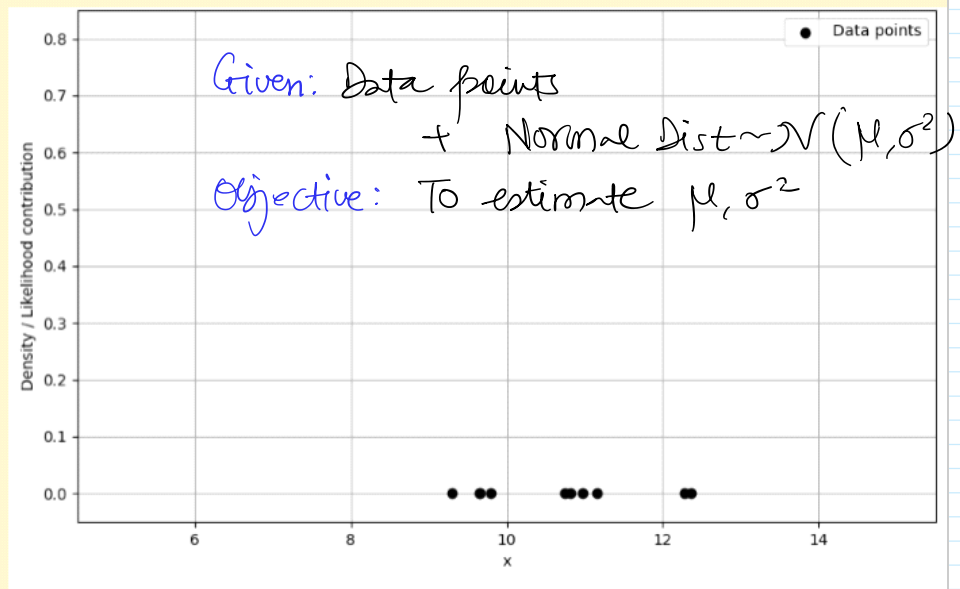
$$\left\{ \begin{matrix} MoM \end{matrix} \right\} \Rightarrow \begin{matrix} \beta \approx 0.77 \\ \alpha \approx \end{matrix}$$

- Equate them to solve for the parameters

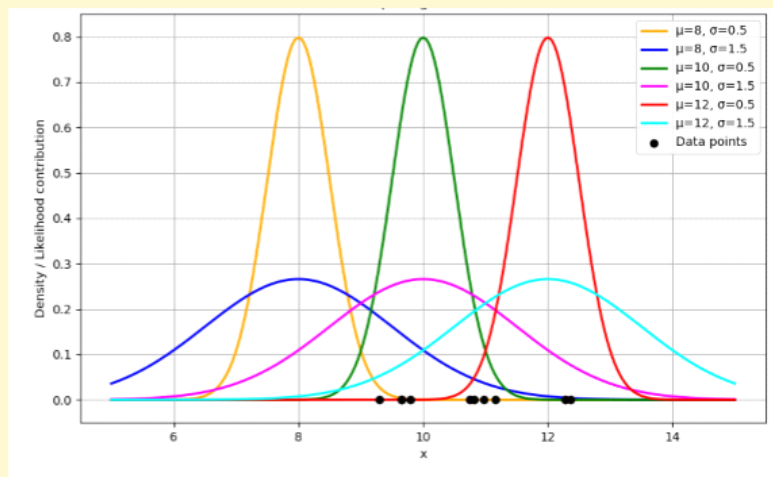
Maximum likelihood estimation

Basic idea

Objective of MLE:



Maximum likelihood estimation



Maximum likelihood estimation

Find $\mathcal{L}(\mu, \sigma | x_i) = f(x_i | \mu, \sigma)$ for each dart, x_i .

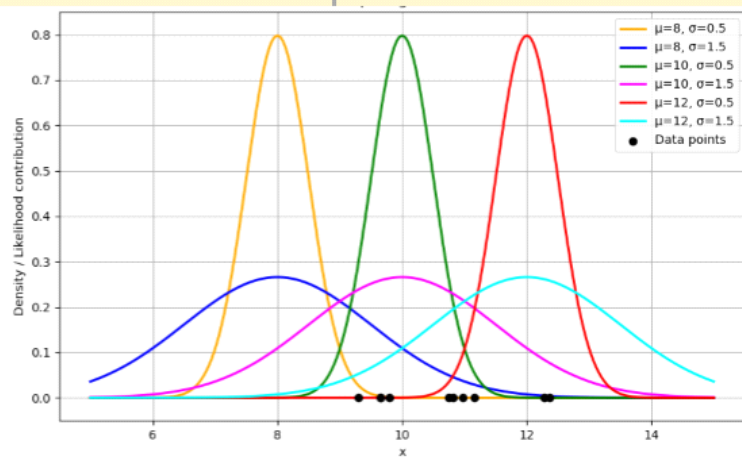
Maximize the product of likelihood,

$$\mathcal{L}(\mu, \sigma | \mathbf{x}) = \prod_i^n f(x_i | \mu, \sigma)$$

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \mathcal{L}(\mu, \sigma | \mathbf{x})$$

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \ln[\mathcal{L}(\mu, \sigma | \mathbf{x})]$$

Excel Demo!



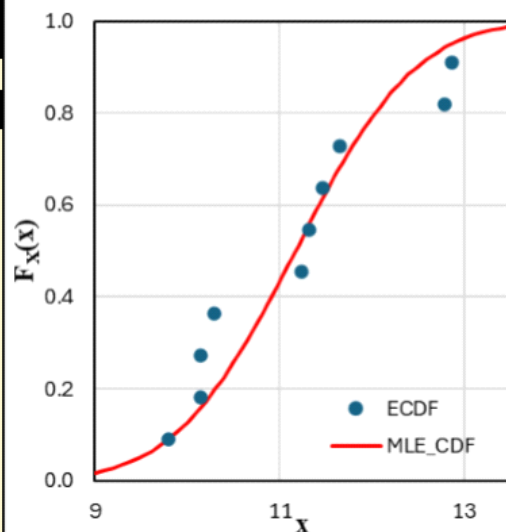
color	oran.	blue	green	mag.	red	turq.
μ_{guess}	8	8	10	10	12	12
σ_{guess}	0.5	1.5	0.5	1.5	0.5	1.5
x_i	Likelihood					
10.7	0.00	0.05	0.26	0.24	0.03	0.19
9.8	0.00	0.13	0.73	0.26	0.00	0.09
11.0	0.00	0.04	0.12	0.22	0.10	0.21
12.3	0.00	0.00	0.00	0.08	0.68	0.26
9.6	0.00	0.15	0.62	0.26	0.00	0.08
9.6	0.00	0.15	0.62	0.26	0.00	0.08
12.4	0.00	0.00	0.00	0.08	0.61	0.26
11.2	0.00	0.03	0.06	0.20	0.19	0.23
9.3	0.03	0.18	0.30	0.24	0.00	0.05
10.8	0.00	0.05	0.21	0.23	0.05	0.19
Sum-log-lik.	-166	-31	-32	-17	-59	-20

$\mu_{\text{goal-seek}}$ **10.67**
 $\sigma_{\text{goal-seek}}$ **1.03**

Lik.	Log-lik.
0.39	-0.4
0.27	-0.6
0.37	-0.4
0.11	-0.9
0.24	-0.6
0.24	-0.6
0.10	-1.0
0.35	-0.5
0.16	-0.8
0.38	-0.4

Use "solver" to **maximize** Sum-log-lik. by changing μ, σ .

Sum-log-lik. **-6.3**



M2C2

Interval Estimation: two-sided and one-sided CI for population μ and σ^2 under known and unknown variance assumptions.

Hypothesis Testing Principles: H_0 and H_1 .

Error Analysis & Test Power: Type I (α) and Type II (β) error probabilities, p -values.

Parametric Testing: Z -tests, t -tests, χ^2 -tests, and F -tests for single-sample and two-sample population parameters.

Empirical Probability Plotting: probability paper and ECDFs to evaluate candidate dist. visually.

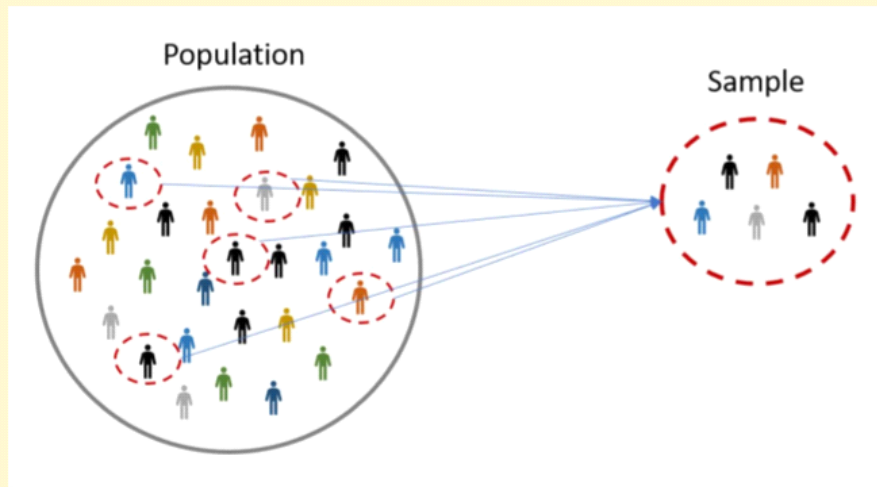
Goodness-of-Fit Verification: χ^2 , Kolmogorov-Smirnov (K-S), and Anderson-Darling (A-D) non-parametric hypothesis tests to validate distribution fits.

Motivation

How to test hypotheses and validate probability models?

Point Estimates $\Rightarrow$ Confidence Interval

- Issue with the point estimate, $\hat{\mu}$



18

Distribution of Mean

- Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables having a distribution with mean μ and var. σ^2 . If Y is the sample average of X_i 's.

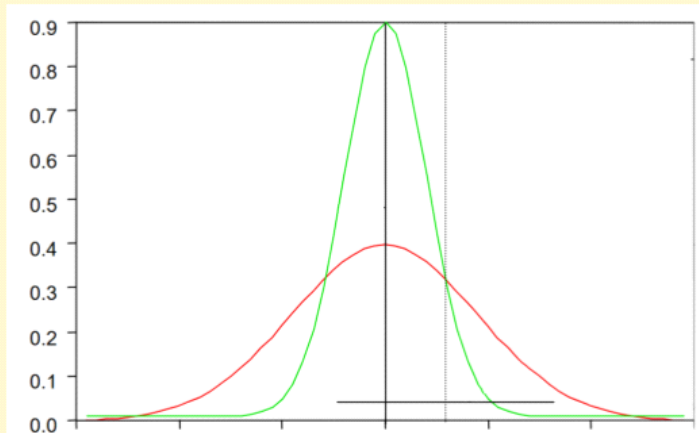
Find μ_Y and σ_Y^2

Confidence interval

- Basic idea:

Confidence interval for μ

- Basic idea:



Confidence interval for μ

- $\bar{x}$:
- α :
- n :
- $100(1 - \alpha)\%$:

Example-1 on CI

- $n = 10$ measurements of impact energy (J) are as follows:
 - 64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, 64.3
 - The sample mean is $\bar{x} = 64.46$
- Assume impact energy is normally distributed with $\sigma = 1$ J
- Find a 95% CI for μ , the mean impact energy

Standard Normal Probabilities

Z-table

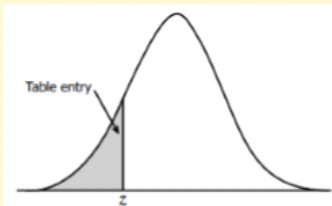
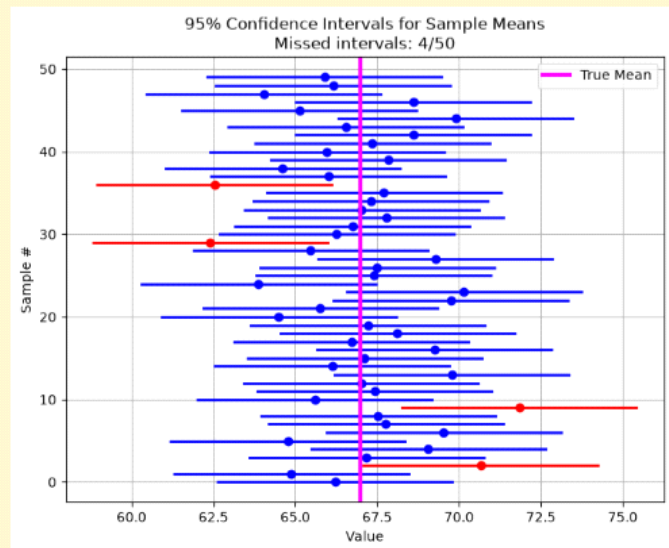


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

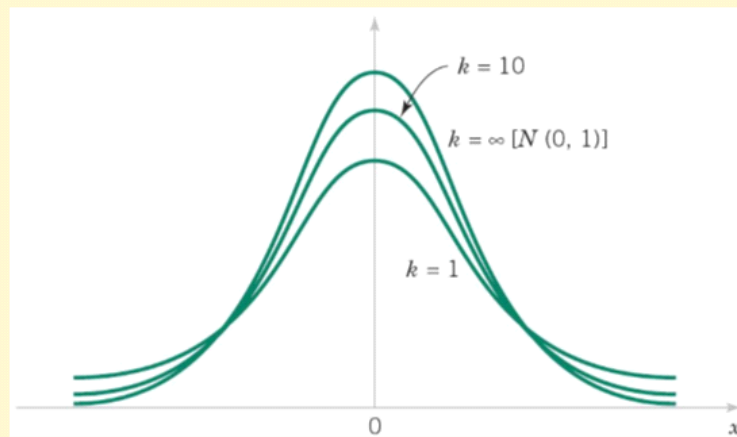
Interpretation of a CI



Choice of sample size

- For a target error in the length of interval

Small sample sizes



27

Thank you!